

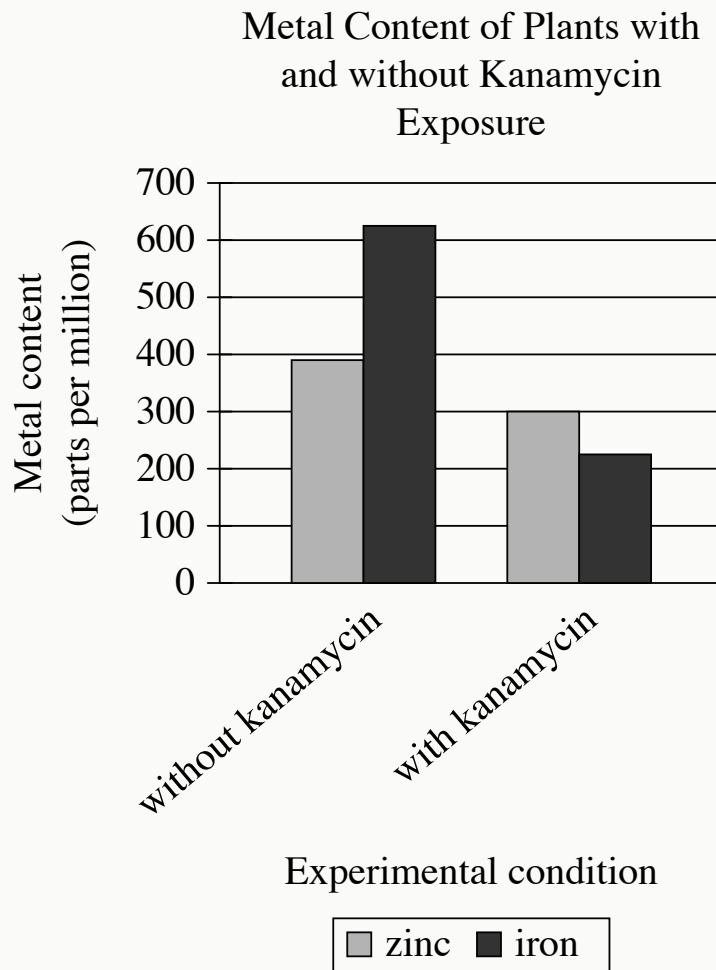
● MEDIUM

● INFORMATION AND IDEAS

Command of Evidence

04

10 questions · ~15 min



- For each data category, the following bars are shown:
 - Zinc
 - Iron
- The data for the 2 categories are as follows:
 - Without kanamycin:
 - Zinc: 390
 - Iron: 625
 - With kanamycin:
 - Zinc: 300
 - Iron: 225

Many plants lose their leaf color when exposed to kanamycin, an antibiotic produced by some soil microorganisms. Spelman College biologist Mentewab Ayalew and her colleagues hypothesized that plants' response to kanamycin exposure involves altering their uptake of metals, such as iron and zinc. The researchers grew two groups of seedlings of the plant *Arabidopsis thaliana*, half of which were exposed to kanamycin and half of which were a control group without exposure to kanamycin, and measured the plants' metal content five days after germination.

Which choice best describes data in the graph that support Ayalew and her colleagues' hypothesis?

- A. The control plants contained higher levels of zinc than iron, but plants exposed to kanamycin contained higher levels of iron than zinc.
- B. Both groups of plants contained more than 200 parts per million of both iron and zinc.
- C. Zinc levels were around 300 parts per million in the control plants but nearly 400 parts per million in the plants exposed to kanamycin.
- D. The plants exposed to kanamycin showed lower levels of iron and zinc than the control plants did.

Survey Results for Two Online Account Sign-in Methods

Sign-in method	Percent of participants in the UK who chose method	Percent of participants in Japan who chose method	Percent of participants in India who chose method
Biometrics (for example, a face scan)	33	29	22
Onetime passcodes	16	8	25

A survey listed methods for signing into online accounts. Participants in different countries were asked to choose the sign-in method they view as most secure. The table presents data for two of the methods. According to the table, onetime passcodes were viewed as most secure by _____ blank

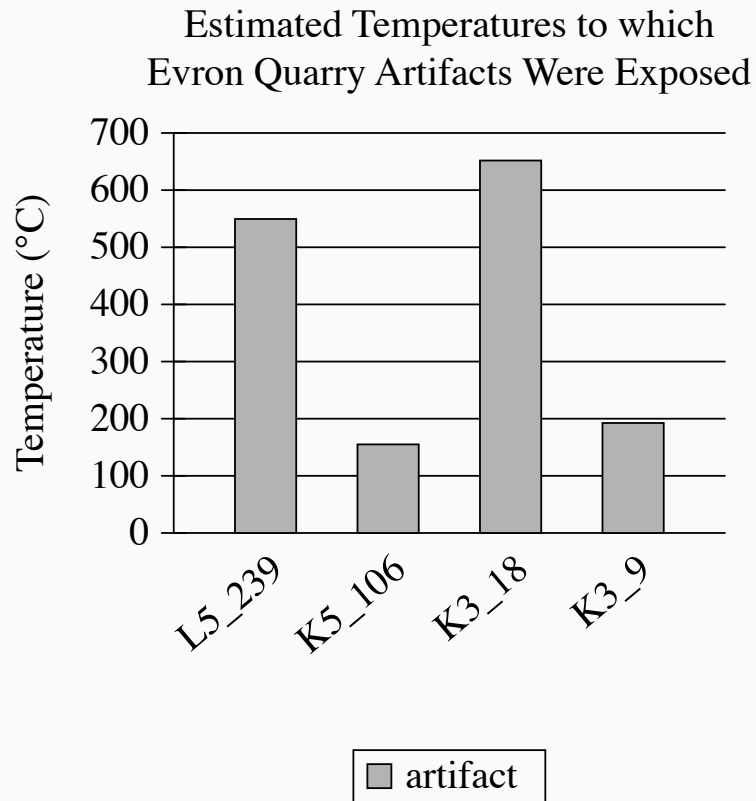
Which choice most effectively uses data from the table to complete the text?

- A. 33 percent of survey participants in the UK.
- B. 22 percent of survey participants in India.
- C. 8 percent of survey participants in Japan.
- D. 16 percent of survey participants in India.

Accomplished printmaker and sculptor Elizabeth Catlett (1915–2012) used her art to explore the Black experience in the United States. In a paper for an art history class, a student claims that Catlett had a particular talent for unifying various artistic traditions and styles in her work.

Which quotation from a scholar describing Catlett’s work would best support the student’s claim?

- A. “In *Mother and Child*, a sculpture of two Black figures, Catlett used an ancient Indigenous sculpting technique and combined the visual aesthetic of modern Mexican muralists with that of German artist Kathe Kollwitz.”
- B. “In her collage *New Generation*, Catlett overlaid fabric onto the canvas to represent the clothing of a father and his toddler, positioned to evoke classic images of a mother and child.”
- C. “Created in 1968, Catlett’s sculpture *Black Unity*, a stylized fist sculpted from mahogany and measuring two feet across, remains an important piece and has received renewed and well-deserved attention in recent years.”
- D. “One series of Catlett’s prints, made by the artist using the linoleum cut method, depicts several notable African American women, including Harriet Tubman and Sojourner Truth.”



Flint artifacts dating to 800,000 to 1,000,000 years ago have been recovered from the Evron Quarry in Israel. Likely created by the hominin *Homo erectus*, the artifacts have no visual features suggesting that they were exposed to fire, leading some scholars to conclude that these hominins had not acquired control of fire. But Zane Stepka and colleagues recently used a new method to determine whether these artifacts had been exposed to temperatures above 400°C (the typical temperature campfires reach) and concluded that the hominins who inhabited the site may have had control of fire.

Which choice best describes data in the graph that support the team's conclusion?

- A. Artifacts K5_106 and K3_9 were exposed to temperatures above 400°C.

- B.** Artifacts L5_239 and K3_18 were exposed to temperatures of approximately 550°C and 650°C, respectively.
- C.** All of the artifacts were exposed to temperatures above 100°C.
- D.** Artifact K3_9 was exposed to a higher temperature than was artifact K5_106.

“Often Rebuked, Yet Always Back Returning” is an 1846 poem by Emily Brontë. The poem conveys the speaker’s determination to experience the countryside around her: _____ blank

Which quotation from the poem most effectively illustrates the claim?

- A. “Often rebuked, yet always back returning / To those first feelings that were born with me, / And leaving busy chase of wealth and learning / For idle dreams of things which cannot be.”
- B. “I’ll walk, but not in old heroic traces, / And not in paths of high morality, / And not among the half-distinguished faces, / The clouded forms of long-past history.”
- C. “I’ll walk where my own nature would be leading: / It vexes me to choose another guide: / Where the grey flocks in ferny glens are feeding; / Where the wild wind blows on the mountain side.”
- D. “To-day, I will seek not the shadowy region; / Its unsustaining vastness waxes drear; / And visions rising, legion after legion, / Bring the unreal world too strangely near.”

Five of the Responses to Survey about Actions to Conserve Energy

Action	Action category	Percentage of respondents selecting action (%)
Use efficient cars/hybrids	efficiency	2.8
Change thermostat setting	curtailment	6.3
Use bike or public transportation instead of car	curtailment	12.9
Use efficient light bulbs	efficiency	3.6
Turn off lights	curtailment	19.6

In a survey of public perceptions of energy use, researcher Shahzeen Attari and her team asked respondents to name the most effective action ordinary people can take to conserve energy. The team categorized each action as either an efficiency or a curtailment and found that respondents tended to name curtailments more often than they did efficiencies. For example, 19.6% of respondents stated that the most effective way to conserve energy is to turn off the lights, while only _____blank

Which choice most effectively uses data from the table to complete the text?

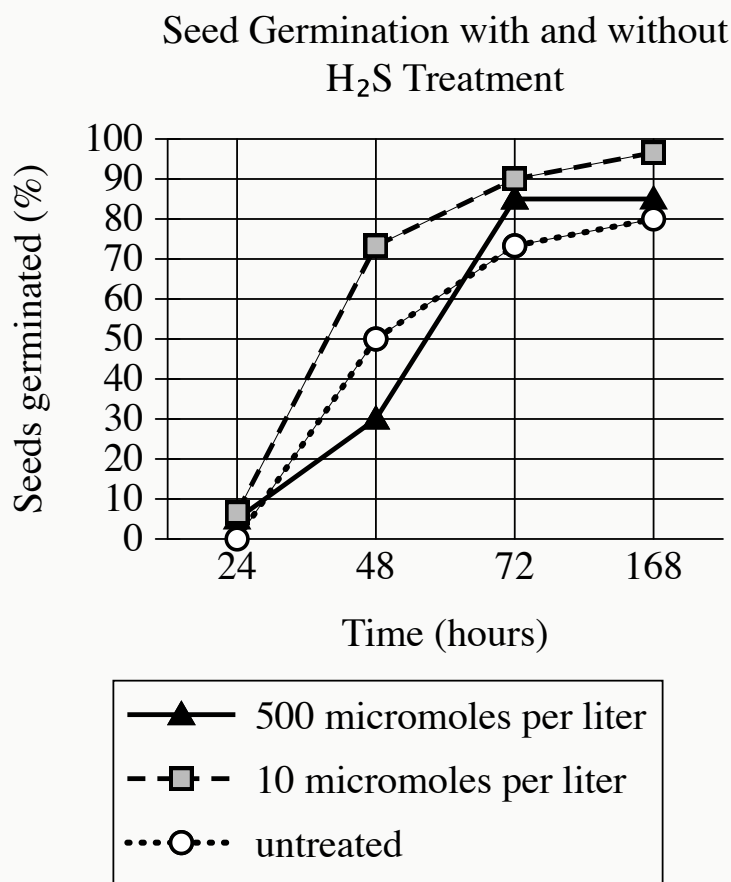
- A. 6.3% of respondents said it was most effective to use efficient cars or hybrids.
- B. 2.8% of respondents said it was most effective to change the thermostat setting.

- C. 12.9% of respondents said it was most effective to use a bike or public transportation.
- D. 3.6% of respondents said it was most effective to use efficient light bulbs.

“Ghosts of the Old Year” is an early 1900s poem by James Weldon Johnson. In the poem, the speaker describes experiencing an ongoing cycle of anticipation followed by regretful reflection: _____blank

Which quotation from “Ghosts of the Old Year” most effectively illustrates the claim?

- A. “What does this brazen tongue declare, / That falling on the midnight air / Brings to my heart a sense of care / Akin to fright?”
- B. “The snow has ceased its fluttering flight, / The wind sunk to a whisper light, / An ominous stillness fills the night, / A pause—a hush.”
- C. “It tells of many a squandered day, / Of slighted gems and treasured clay, / Of precious stores not laid away, / Of fields unreaped.”
- D. “And so the years go swiftly by, / Each, coming, brings ambitions high, / And each, departing, leaves a sigh / Linked to the past.”



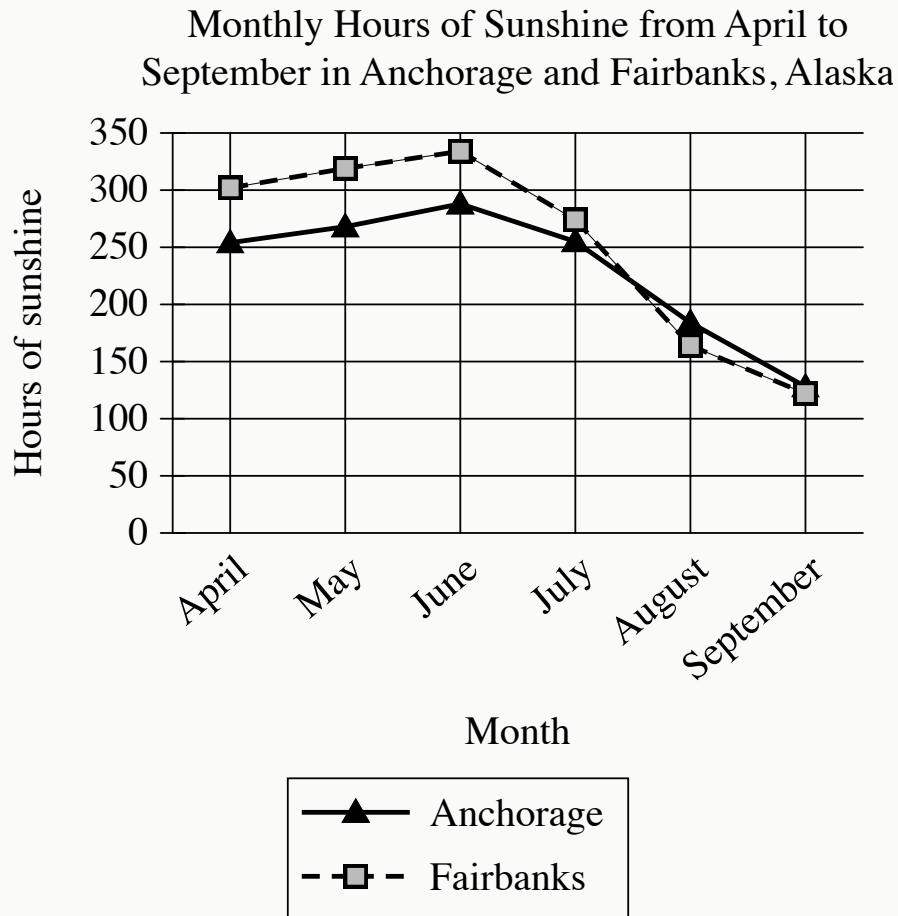
In high concentrations, hydrogen sulfide (H₂S) is typically toxic to many plants. Frederick D. Dooley and colleagues wanted to understand what effects low doses of H₂S might have on plant growth. They treated bean, corn, wheat, and pea seeds with various concentrations (measured in micromoles per liter) of H₂S and tracked the germination of those seeds along with the germination of untreated seeds. Treatment with particular concentrations of H₂S was associated with accelerated germination: for example, _____blank

Which choice most effectively uses data from the graph to complete the statement?

- A. at 24 hours, less than 10% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas more than 90% of those seeds had

germinated at 168 hours.

- B.** at 48 hours, more than 70% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 50% of untreated seeds had germinated.
- C.** at 168 hours, more than 90% of seeds treated with H₂S at concentrations of 10 or 500 micromoles per liter had germinated, whereas less than 70% of untreated seeds had germinated.
- D.** at 48 hours, approximately 50% of seeds treated with H₂S at a concentration of 10 micromoles per liter had germinated, whereas only approximately 30% of untreated seeds had germinated.



- The following 2 lines are shown:
 - Anchorage
 - Fairbanks
- The Anchorage line:
 - Begins at April, 254 hours
 - Rises gradually to May, 268 hours
 - Rises gradually to June, 288 hours
 - Falls gradually to July, 255 hours
 - Falls sharply to August, 184 hours
 - Falls sharply to September, 128 hours
- The Fairbanks line:
 - Begins at April, 302 hours

- Rises gradually to May, 319 hours
- Rises gradually to June, 334 hours
- Falls sharply to July, 274 hours
- Falls sharply to August, 164 hours
- Falls sharply to September, 122 hours

A student is researching monthly hours of sunshine in different cities in Alaska. When comparing trends in Anchorage and Fairbanks, the student concludes that the two cities show a similar pattern in the monthly hours of sunshine from April to September.

Which choice best describes data from the graph that support the student's conclusion?

- A. The monthly hours of sunshine in both Anchorage and Fairbanks hold steady in June and July before beginning to decline in August.
- B. The monthly hours of sunshine in both Anchorage and Fairbanks increase from April to June and then decrease from June to September.
- C. Anchorage and Fairbanks both have less than 200 monthly hours of sunshine from April to September.
- D. Anchorage and Fairbanks both have more than 300 monthly hours of sunshine from April to June and less than 200 hours from July to September.

“Ad recall” measures how memorable an advertising campaign is. To provide advertisers with information about their ads’ memorability, a social media site regularly surveys users about whether they remember ads they had recently interacted with on the site. In a study that drew on this survey data, advertising researcher Kristen Sussman and colleagues noted that different kinds of social media interactions involve different levels of cognitive engagement: commenting on or sharing a post is more cognitively demanding than is clicking on embedded links or on a “like” button. The researchers hypothesized that interactions indicating high levels of cognitive engagement with ad content would result in relatively high levels of ad recall.

Which finding, if true, would most directly support the researchers’ hypothesis?

- A. Users who interacted with an ad were much more likely to do so by clicking on the ad’s “like” button than they were to interact with the ad in any other way.
- B. Users who interacted with an ad were significantly more likely to purchase the advertised product at the time they saw the ad than were users who saw the ad but did not interact with it.
- C. Compared with users who clicked on links in an ad, users who commented on that same ad were significantly more likely to remember seeing the ad when surveyed two days later.
- D. Although users who shared an ad were highly likely to remember details from the ad when surveyed two days later, those same users tended to forget those details when surveyed again a week later.